MUHAMMAD MUZAMMIL

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SKILLS

AI & Generative AI Engineering:

- **LLMs:** OpenAI GPT-4o, Anthropic Claude 3.5 Sonnet, Llama 3, Mistral etc.
- **RAG Architectures:** Semantic search, hybrid search, reranking pipelines, parallel ingestion pipeline, prompt engineering etc.
- **Vector Databases:** Pinecone, Chroma, Qdrant, pgvector
- **Frameworks:** LangChain, LlamaIndex, LangGraph, AWS Strands, Google Agentic Development Kit
- **Agentic AI:** Autonomous agents, multi-agent orchestration, tool calling, self-correction loops

Spec-Drive AI Development (Vibe Coding)

- **AI Code Companions:** Cursor IDE, Claude Code, GitHub Copilot
- **Workflow Optimization:** Prompt engineering, structured system prompts, context window management
- **Rapid Prototyping:** High-velocity microservices development, AI-driven code generation

Backend Engineering

- **Core Languages:** Python (Expert), JavaScript, HTML/CSS
- **Frameworks:** FastAPI, Django REST Framework (DRF), Flask
- **Architecture:** Microservices, RESTful APIs, Asynchronous Programming, Event Driven Architecture
- **Databases:** PostgreSQL, PyMongo (MongoDB), SQL optimization, schema design

MLOps & DevOps

- **Containerization & Orchestration:** Docker, Kubernetes
- **Cloud Infrastructure:** AWS (EC2, S3, RDS, Lambda), Azure

WORK EXPERIENCE

Sr. Backend & AI Engineer

VentureDive | May 2024 - Present

- Develop AI-powered, scalable applications and production-ready Gen-AI systems (LLMs, RAG).
- Optimised a large-scale RAG system managing 90M+ embeddings, reducing query retrieval latency from 15+ minutes to 2–3 seconds by implementing BM25 hybrid indexing, tuning distance metric algorithms, and optimizing query execution.
- Engineered autonomous Agentic AI systems, including a production-ready Voice-to-Voice AI Agent integrated with Twilio telephony for real-time mobile calling, tool use, and automated task execution.
- Adopted Vibe Coding methodologies to maintain and scale a legacy Node.js AI application; successfully resolved complex bugs and shipped new production features through high-velocity, spec-driven AI development without prior framework expertise.
- Containerize legacy projects and optimize code architecture to meet SonarQube quality standards.
- Build highly secure, high-performance microservices and APIs using FastAPI and Django REST Framework (DRF).

Backend Developer

Techfy | Aug 2023 - April 2024

- Architected and built an automated PaaS (Platform as a Service) infrastructure on AWS and Kubernetes to automatically containerize, isolate, and deploy user-submitted Odoo source code via secure CI/CD pipelines and automated Ingress routing.
- Engineered and delivered end-to-end software solutions seamlessly to clients.
- Designed and deployed robust DevOps infrastructure leveraging AWS, Kubernetes, and CI/CD pipelines.
- Managed, optimized, and enhanced the performance and security of existing web applications and microservices.
- Implemented secure coding practices across the entire product ecosystem.

Python Developer

JagahOnline | Jan 2022 - July 2023

- Architected custom CRM web applications utilizing Django.
- Developed high-performance microservices for mobile applications using FastAPI.
- Built advanced data scraping frameworks and automation pipelines via Selenium and BeautifulSoup.
- Executed complex SQL queries and engineered optimized database schemas for smooth operations.
- Conducted rigorous data cleaning, preprocessing, and analysis utilizing Pandas.

- **CI/CD & Testing:** Automated deployment workflows, SonarQube code quality gating
- **Automation:** Selenium, BeautifulSoup (bs4), web scraping pipelines

Computer Vision & Deep Learning

- **Frameworks:** TensorFlow, Keras, PyTorch
- **Core Vision:** OpenCV, object detection (YOLO), image classification (ResNet50)
- **GUI Development:** Tkinter, PyQt5

EDUCATION

- BS Data Science - Ziauddin University FEST (2022-2026)

INTERNATIONAL CERTIFICATIONS

- DeepLearning.AI TensorFlow Developer
- Python for Data Science, AI & Development
- Python for Cybersecurity

OPEN SOURCE CONTRIBUTIONS

Core Contributor — LlamaIndex Vector Store Ecosystem:

- Developed and submitted an upstream pull request expanding database retrieval capabilities within LlamaIndex.
- Designed custom abstraction classes to integrate multi-metric vector distance algorithms (e.g., Euclidean, Hamming) supported natively by pgvector, bypassing the framework's previous Cosine-only limitation.

Core Contributor — VannaAI (Ollama Integration):

- Authored and submitted a feature patch to the VannaAI repository to optimize the UI rendering pipeline for reasoning/thinking LLMs.
- Introduced a configurable initialization parameter (thinking = True/False) within the Ollama integration module, enabling developers to cleanly toggle thought-process token streaming and prevent UI breaks during data generation.

Computer Vision & Deep Learning Engineer

Bleed AI | Nov 2021 - Feb 2022

- Developed and deployed robust deep learning and computer vision architectures.
- Collaborated directly with clients to translate business requirements into technical implementations.
- Optimized existing models to maximize performance, speed, and accuracy.
- Designed a comprehensive product showcase gallery featuring a diverse portfolio of image processing projects.

AI Developer

GrainScan | Sep 2020 - Oct 2021

- Developed Pakistan's first AI-powered Rice Quality Analyzer desktop application.
- Built a highly scalable, Python-based desktop application framework.
- Led product R&D to continuously integrate innovative machine learning features.
- Designed interactive, user-friendly desktop GUIs utilizing Tkinter and PyQt5.
- Trained deep learning architectures on ResNet50 and maximized accuracy via hyperparameter tuning.
- Automated ML pipelines and computer vision deployment workflows.

Research Intern

Smart City Lab, NCAI, NED University Karachi | Nov 2019 - Jan 2020

- Hands on learning on Python, Computer Vision, ML & AI
- Contributed in preparation of data sets for multiple projects